C.W

Recursive Definition

07/09/2026

- Recursion - ending condition is a must.

(*) $f(0) = 3$; find $f(4) = ?$

$f(n+1) = 2f(n) + 3$

Solution:

- $f(4) = f(3+1) = 2 \cdot f(3) + 3$

$\therefore f(3) = f(2+1) = 2 \cdot f(2) + 3$

$\therefore f(2) = f(1+1) = 2 \cdot f(1) + 3$

$\therefore f(1) = f(0+1) = 2 \cdot f(0) + 3$

$= 2 \cdot 3 + 3$

$= 9$

$\therefore f(2) = 2 \cdot 9 + 3 = 21$

$\therefore f(3) = 2 \cdot 21 + 3 = 45$

$\therefore f(4) = 2 \cdot 45 + 3$

$= 93$ (Ans)

(*) $F(0)=1$; find $F(5)=?$

$F(n+1) = (n+1)F(n)$

$\Rightarrow F(5) = F(4+1) = (4+1)F(4)$

$F(4) = (3+1)F(3)$

$F(3) = (2+1)F(2)$

$F(2) = (1+1)F(1)$

$F(1) = (0+1)F(0)$

$= 1 \cdot 1 = 1$

$\therefore F(2) = 2 \cdot 1 = 2$

$\therefore F(3) = 3 \cdot 2 = 6$

$\therefore F(4) = 4 \cdot 6 = 24$

$\therefore F(5) = 5 \cdot 24 = 120$

(A)

(*) (1) if $m=0$ then $A(m,n) = n+1$

(2) if $m \neq 0$ but $n=0$ then $A(m,n)$

$= A(m-1, 1)$

(3) if $m \neq 0$ and $n \neq 0$ then $A(m,n)$

$= A(m-1, A(m, n-1))$

Find $A(1,3) = ?$

Solution:

[Ackermann function এর]

[উপরের 3টি condition এই function এর definition. এই Def অনুসরণ করতে হবে.]

$A(1,3) = A(0, A(1,2))$

$A(1,2) = A(0, A(1,1))$

$A(1,1) = A(0, A(1,0))$

$A(1,0) = A(0,1)$

$A(0,1) = 1+1 = 2$

$\therefore A(1,0) = 2$

$\therefore A(1,1) = A(0,2)$

$= 2+1 = 3$

$\therefore A(1,2) = A(0,3)$

$= 3+1 = 4$

$\therefore A(1,3) = A(0,4)$

$= 4+1 = 5$ (A)

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H.W:

$f_0(0) = 0$, $f(1) = 1$

and ; find $f(7) = ?$

$f(n) = f(n-1) + f(n-2)$